

Inverse Kinematics Wrap-Up and Intro to Jacobians

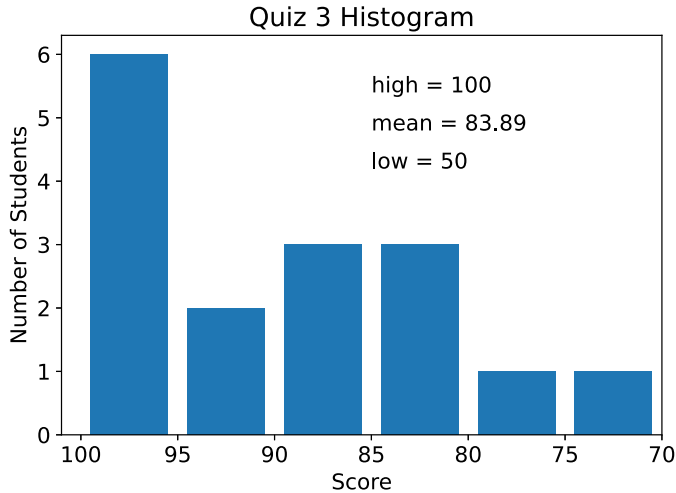
EGR 445/545

GVSU

Plan for Today

- wrap-up inverse kinematics
 - quiz 3 solution
 - lingering questions
- intro to Jacobians
 - what is a Jacobian and why should you care?
 - how are Jacobians used and what can they do for us?
 - what is a singularity and how do we find them?

Quiz 3 Histogram



Intro to Jacobians

Intro to Jacobians: Key Questions

- what is a singularity?
 - how do singularities affect robot performance?
 - how do you find the singularities of a robot?
- what is a Jacobian?
 - how are Jacobians used in robotics?
 - how do Jacobians help us find singularities?
- how would you find the velocity of the tip of the robot?
 - why would you want to find the tip velocity?

Why do Jacobians and Singularities matter?

? How do robots move in straight lines or in world coordinates?

→ How should the servos move to cause desired tip velocity?

A Jacobian is a matrix of partial derivatives

In robotics

$$\begin{bmatrix} v_{tipx} \\ v_{tipy} \\ v_{tipz} \\ \omega_{tipx} \\ \omega_{tipy} \\ \omega_{tipz} \end{bmatrix}$$

=

$$\begin{bmatrix} J \end{bmatrix}_{6 \times 6}$$

$$\begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \\ \dot{\theta}_4 \\ \dot{\theta}_5 \\ \dot{\theta}_6 \end{bmatrix}$$

How do I solve for the joint velocities that I need to generate my desired tip velocity?

$$\begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \vdots \\ \dot{\theta}_6 \end{bmatrix} = [J]^{-1} \begin{bmatrix} v_{TIPx} \\ v_{TIpy} \\ v_{TIPz} \\ \omega_x \\ \omega_y \\ \omega_z \end{bmatrix}$$

desired tip velocity

What happens if J^{-1} does not exist?

→ if determinant of a matrix $= 0$, it is not invertible

→ equivalently, if a matrix is not full rank, it is not invertible

A position where the determinant of the Jacobian of a robot is zero is called a singularity.

Chapter 5 Reading Plan

- read sections 5.1-5.3
 - more than skimming, but don't get too bogged down
- skip 5.4
- 5.5-5.8 is the heart of the chapter
 - read carefully
 - understanding equations 5.45 and 5.47 is key
- skim 5.9 and 5.10
- skip 5.11

Big Idea
→ work our way out to tip to find

N $\overline{V_{tip}}$ and

N $\overline{w_{link N}}$

Chapter 5 Learning Plan

The main things I want you to learn from chapter 5:

- how to find the tip velocity and the rotational velocity of the final link based on equations 5.45 and 5.47
 - also knowing how to substitute in eqns. 5.48 for prismatic joints
- how to find a Jacobian
- how to find singularities from a Jacobian

What is a Singularity?

- practically: a configuration where a robot loses one or more degrees of freedom
- mathematically: a configuration where the determinant of the Jacobian is zero

What is a Jacobian?

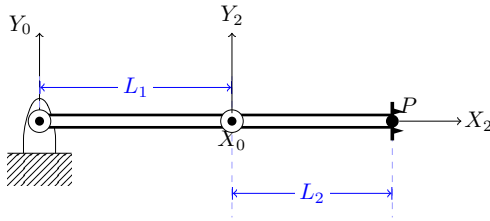
- a matrix of partial derivatives that maps from the joint velocities to the tip velocities

$$\begin{bmatrix} v_{tipx} \\ v_{tipy} \\ v_{tipz} \\ \omega_{linkNx} \\ \omega_{linkNy} \\ \omega_{linkNz} \end{bmatrix} = [\mathbf{J}] \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \\ \dot{\theta}_4 \\ \dot{\theta}_5 \\ \dot{\theta}_6 \end{bmatrix} \quad [\mathbf{J}] = \begin{bmatrix} \frac{\partial v_{tipx}}{\partial \dot{\theta}_1} & \frac{\partial v_{tipx}}{\partial \dot{\theta}_2} & \dots \\ \frac{\partial v_{tipy}}{\partial \dot{\theta}_1} & \frac{\partial v_{tipy}}{\partial \dot{\theta}_2} & \dots \\ \vdots & \vdots & \\ \frac{\partial \omega_{linkNz}}{\partial \dot{\theta}_1} & \frac{\partial \omega_{linkNz}}{\partial \dot{\theta}_2} & \dots \end{bmatrix}$$

- $\bar{v}_{tip}$ is the tip velocity in $\hat{X}_0$, $\hat{Y}_0$, and $\hat{Z}_0$ directions
- $\bar{\omega}_{linkN}$ is the angular or rotational velocity of the final link about the $\hat{X}_0$, $\hat{Y}_0$, and $\hat{Z}_0$ axes

Quick Example: Two Link Robot

Find the Jacobian of the two link robot in the configuration shown below:



- find $\bar{v}_{tip}$
 - use $\bar{v} = \bar{\omega} \times \bar{r}$
- take the partial derivatives of $\bar{v}_{tip}$ w.r.t. the joint angles
- is this configuration a singularity?

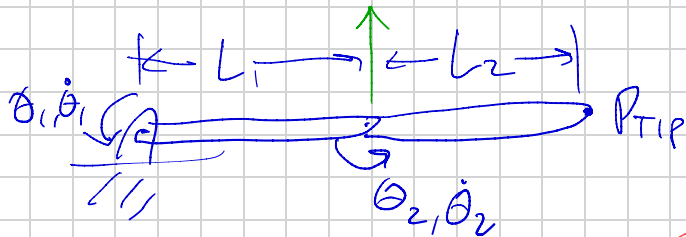
- the Jacobian of a robot depends on the position / pose / joint angles

→ In order to find the Jacobian, we need a symbolic expression for $\bar{v}_{tip}$ and $\bar{\omega}_{tip}$

Two link example

Assume $\theta_1 = \theta_2 = 0$
for now

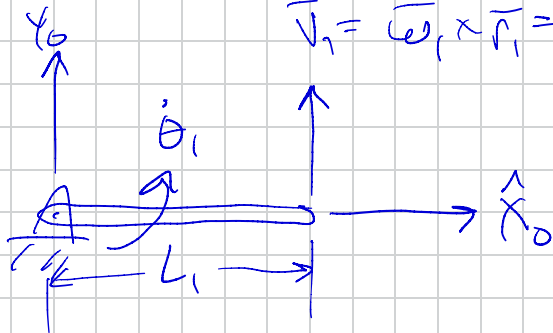
$$\begin{bmatrix} v_{TIPx} \\ v_{TIpy} \end{bmatrix} = [J] \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}$$



$$\bar{v}_{tip} = ?$$

$$\bar{v} = \bar{\omega} \times \bar{r}$$

Start with 1st link



$$\bar{V}_1 = \bar{\omega}_1 \times \bar{r}_1 = \begin{bmatrix} 0 \\ \dot{\theta}_1 L_1 \\ 0 \end{bmatrix}$$

$$\bar{r}_1 = \begin{bmatrix} L_1 \\ 0 \\ 0 \end{bmatrix}$$

$$\bar{\omega}_1 = \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix}$$

? $\begin{pmatrix} i & j & k \\ i & j & k \end{pmatrix}$

$$\begin{array}{c} \xrightarrow{\quad} + \\ i \ j \ k \ i \ j \ k \\ \xleftarrow{\quad} - \end{array}$$

$$\hat{k} \times \hat{i} = +\hat{j}$$

Link 2:

eqn 5.46

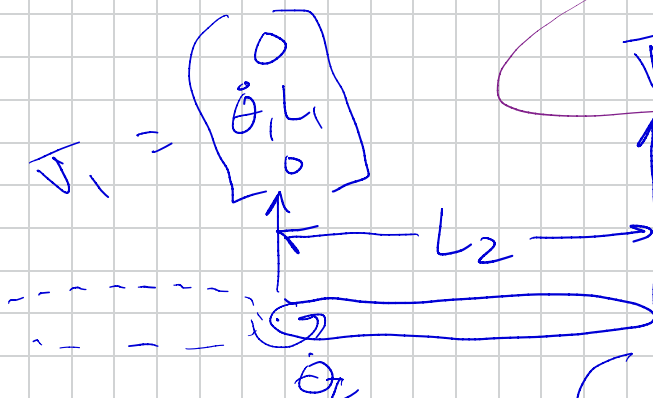


Diagram illustrating the velocity of Link 2 relative to Link 1. The velocity vector $\bar{V}_2$ is the sum of the velocity of the joint $\bar{V}_1$ and the velocity due to the rotation of Link 2 relative to Link 1, $\bar{\omega}_2 \times \bar{r}_2$.

$$\bar{V}_2 = \bar{V}_1 + \bar{\omega}_2 \times \bar{r}_2$$

The velocity vectors are defined as:

$$\bar{V}_1 = \begin{bmatrix} 0 \\ \dot{\theta}_1 L_1 \\ 0 \end{bmatrix}$$

$$\bar{V}_2 = \begin{bmatrix} 0 \\ \dot{\theta}_1 L_1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 + \dot{\theta}_2 \end{bmatrix} \begin{bmatrix} L_2 \\ 0 \\ 0 \end{bmatrix}$$

The resulting velocity vector for Link 2 is:

$$\bar{V}_2 = \begin{bmatrix} 0 \\ \dot{\theta}_1 L_1 + (\dot{\theta}_1 + \dot{\theta}_2) L_2 \\ 0 \end{bmatrix}$$

When $\theta_1 = \theta_2 = 0$

$$V_{TUP} = \begin{bmatrix} 0 & V_{TUPx} \\ \dot{\theta}_1 L_1 + (\dot{\theta}_1 + \dot{\theta}_2) L_2 & V_{TUPy} \\ 0 & \end{bmatrix}$$

$$J = \begin{bmatrix} \frac{\partial V_{TUPx}}{\partial \dot{\theta}_1} & \frac{\partial V_{TUPx}}{\partial \dot{\theta}_2} \\ \frac{\partial V_{TUPy}}{\partial \dot{\theta}_1} & \frac{\partial V_{TUPy}}{\partial \dot{\theta}_2} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ L_1 + L_2 & L_2 \end{bmatrix}$$

$$\dot{\theta}_1 = \dot{\theta}_2 = 0$$



$$J = \begin{bmatrix} 0 & 0 \\ L_1 + L_2 & L_2 \end{bmatrix}$$

The robot tip
cannot move in
the x direction
in this

$$\begin{bmatrix} v_{tipx} \\ v_{tipty} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ L_1 + L_2 & L_2 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}$$

configuration
→ tip must go up
or down

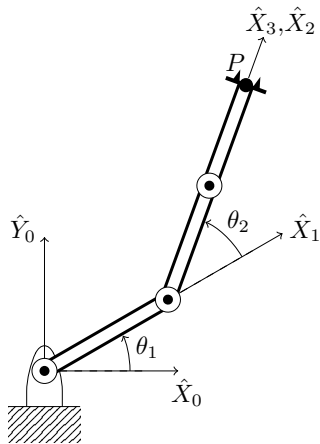
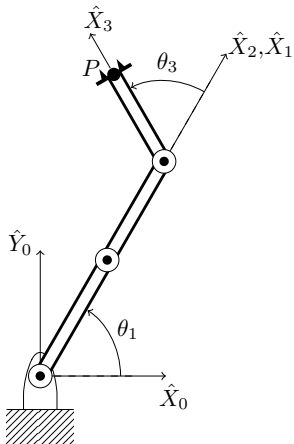
How are Jacobians and Singularities Related?

$$\begin{bmatrix} \dot{\theta}_i \end{bmatrix} = \begin{bmatrix} \mathbf{J} \end{bmatrix}^{-1} \begin{bmatrix} \bar{v}_{tip} \\ \bar{\omega}_{linkN} \end{bmatrix}$$

- a robot is in a singularity configuration when the Jacobian is not invertible
 - practically: you cannot freely choose the tip velocity
 - the robot has lost at least one degree of freedom
 - mathematically: a matrix is not invertible when its determinant is zero or it has less than full rank
 - those are two ways of saying the same thing

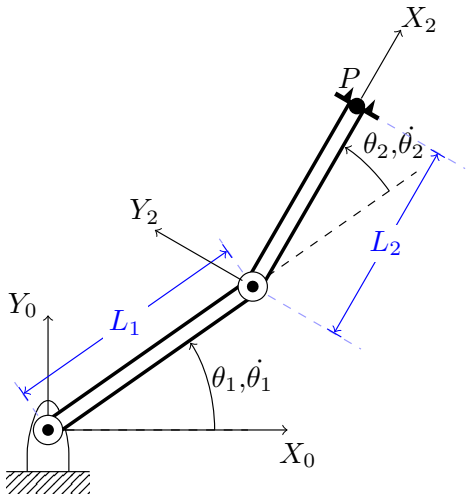
We must do the math

One of these is a in a singularity and the other is not:



General Two Link Jacobian

- how do we find $\bar{v}_{tip}$ in general?
- $\bar{\omega}$ and $\bar{r}$ must be in the same frame for cross products



Finding the tip velocity and omega for the last link

- how do we find v_{tip} and ω_N symbolically?

Velocity propagation

- how do we handle velocity in a moving reference frame?

Sympy Learning Objectives

- know how to use `sympy` to find Jacobians and singularity conditions
 - use eqns. 5.45-5.47 to find v_{tip} and ω_N
 - take partial derivatives to find the Jacobian
 - simplify the determinant to find the singularity conditions

Jacobian Math and Eqns. 5.45-5.47

Velocity Propagation Equations

- understand eqns. 5.45-5.47
- you may have learned $\bar{v} = \bar{\omega} \times \bar{r}$ in dynamics
 - if the reference frame is moving, we have to add that in as well
- eqn. 5.46: ${}^i v_{i+1} = {}^i v_i + {}^i \bar{\omega}_i \times {}^i P_{i+1}$
 - example: ${}^1 v_2 = {}^1 v_1 + {}^1 \bar{\omega}_1 \times {}^1 P_2$
- eqn. 5.47: ${}^{i+1} v_{i+1} = {}^{i+1} \mathbf{R} ({}^i v_i + {}^i \bar{\omega}_i \times {}^i P_{i+1})$
 - example: ${}^2 v_2 = {}^2 \mathbf{R} ({}^1 v_1 + {}^1 \bar{\omega}_1 \times {}^1 P_2) = {}^2 \mathbf{R} {}^1 v_2$
- eqn. 5.45: ${}^{i+1} \bar{\omega}_{i+1} = {}^{i+1} \mathbf{R} {}^i \bar{\omega}_i + \dot{\theta}_{i+1} {}^{i+1} \hat{Z}_{i+1}$

Omega Propagation

- ${}^{i+1}\bar{\omega}_{i+1} = {}_i^{i+1}\mathbf{R} {}^i\bar{\omega}_i + \dot{\theta}_{i+1} {}^{i+1}\hat{Z}_{i+1}$
 - ${}^{i+1}\bar{\omega}_{i+1}$ refers to the **total** angular (rotational) velocity of link $i+1$ in frame $\{i+1\}$
 - includes the contributions of all lower numbered joints
 - ${}^3\bar{\omega}_3$ includes $\dot{\theta}_1$, $\dot{\theta}_2$, and $\dot{\theta}_3$

Omega Propagation cont.

Rotate into frame $i + 1$ and add the $\dot{\theta}_{i+1}$ term:

$${}^{i+1}\bar{\omega}_{i+1} = {}_i^{i+1}\mathbf{R} {}^i\bar{\omega}_i + \dot{\theta}_{i+1} {}^{i+1}\hat{Z}_{i+1}$$

Assuming all joints are rotational:

$${}^1\bar{\omega}_1 = \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix}$$

$${}^2\bar{\omega}_2 = {}_1^2\mathbf{R} {}^1\bar{\omega}_1 + \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_2 \end{bmatrix}$$

• **Note:** ${}_1^2\mathbf{R} = {}_2^1\mathbf{R}^T$; ${}_2^3\mathbf{R} = {}_3^2\mathbf{R}^T$

$${}^3\bar{\omega}_3 = {}_2^3\mathbf{R} {}^2\bar{\omega}_2 + \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_3 \end{bmatrix}$$

Frame Origin Velocities

Note: The origin of frame $\{2\}$ is at the tip of link 1

- O_2 rotates with angular velocity ${}^1\bar{\omega}_1$
- ${}^1v_2 = {}^1v_1 + {}^1\bar{\omega}_1 \times {}^1P_2$
 - velocity of O_2 expressed in frame 1 components
- does 1v_2 depend on $\dot{\theta}_2$?

Frame Origin Velocities

- **Note:** The origin of frame $\{2\}$ is at the tip of link 1
 - the origin for frame $\{3\}$ is at the tip of link 2
 - and so on:

$${}^i v_{i+1} = {}^i v_i + {}^i \bar{\omega}_i \times {}^i P_{i+1}$$

- ${}^1 v_2 = {}^1 v_1 + {}^1 \bar{\omega}_1 \times {}^1 P_2$
- ${}^2 v_3 = {}^2 v_2 + {}^2 \bar{\omega}_2 \times {}^2 P_3$